

Microeconomics Analysis: Search

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Plan

- Fixed sample search
- Sequential search
- Ordered search

1 Fixed Sample Search

2 Sequential Search

3 Ordered Search

FixS: Setup

- There is a (large) number of boxes with unknown prizes
- Each box k contains a prize u_k – a random variable distributed according to $F(u)$
- Prizes are i.i.d.
- Assume that $F(u)$ is continuous, differentiable and has full support
- Assume either finite support $[\underline{u}, \bar{u}]$ or finite moments of the distribution
- Also usually assume $\underline{u} \geq 0$, and 0 outside option
- Pandora has to decide how many boxes she wishes to open
- Opening each box costs s
- After decided number of boxes is opened, Pandora collects the largest prize

FixS: Preliminaries

Suppose that Pandora decides to open n boxes. What is her expected prize?

We need to find the distribution of $\hat{u} = \max\{u_1, \dots, u_n\}$

What is the probability that $\hat{u} \leq u$?

$$\Pr(\hat{u} \leq u) = \Pr(u_1 \leq u, \dots, u_n \leq u) =$$

$$\Pr(u_1 \leq u) \times \dots \times \Pr(u_n \leq u) = [F(u)]^n$$

FixS: Optimal Search

Utility from search:

$$\int_{\underline{u}}^{\bar{u}} u d[F(u)]^n = n \int_{\underline{u}}^{\bar{u}} u [F(u)]^{n-1} f(u) du$$

Search cost:

$$ns$$

We need to compare **marginal effects** from sampling an extra box.

FixS: Optimal Search

Difference between gains from sampling n and $n - 1$ boxes is given by:

$$\begin{aligned}
 g(n) &= \int_{\underline{u}}^{\bar{u}} u d[F(u)]^n - \int_{\underline{u}}^{\bar{u}} u d[F(u)]^{n-1} \\
 &= [uF(u)^n]_{\underline{u}}^{\bar{u}} - \int_{\underline{u}}^{\bar{u}} [F(u)]^n du - [uF(u)^{n-1}]_{\underline{u}}^{\bar{u}} + \int_{\underline{u}}^{\bar{u}} [F(u)]^{n-1} du \\
 &= \bar{u} - \underline{u} + \int_{\underline{u}}^{\bar{u}} [1 - F(u)][F(u)]^{n-1} du \\
 &= \int_{\underline{u}}^{\bar{u}} [1 - F(u)][F(u)]^{n-1} du
 \end{aligned}$$

Thus, the optimal number of boxes is given by

$$s = g(n)$$

We need to check that this such n exists, at the optimal point gains intersect costs from above and that solution is unique.

FixS: Existence and Uniqueness

We treat n and study properties of $g(n)$

$$g(n) = \int_{\underline{u}}^{\bar{u}} [1 - F(u)][F(u)]^{n-1} du$$

Note that $g(n)$ is decreasing in n and thus if solution for $g(n) = s$ exists, it is unique

Moreover,

$$\lim_{n \rightarrow \infty} g(n) = 0$$

and thus the existence of solution is guaranteed if it is worth opening at least one box, i.e. (if $\underline{u} = 0$)

$$s < g(1) = \int_{\underline{u}}^{\bar{u}} u dF(u)$$

FixS: Remark

Let n^* be the solution to $g(n) = s$, $n^* \in \mathbb{R}$.

If $n^* \in \mathbb{N}$, then Pandora is indifferent between opening n and $n - 1$ boxes.

If $n^* \notin \mathbb{N}$, then Pandora prefers to open $n - 1$ boxes.

1 Fixed Sample Search

2 Sequential Search

3 Ordered Search

SeqS: Setup

- There are T of boxes with unknown prizes
- Each box k contains a prize u_k – a random variable distributed according to $F(u)$
- Prizes are i.i.d.
- Assume that $F(u)$ is continuous, differentiable and has full support
- Assume either finite support $[\underline{u}, \bar{u}]$ or finite moments of the distribution
- $\underline{u} \geq 0$, 0 is the outside option
- Opening each box costs s
- Pandora discounts future with discount factor δ per search round
- Pandora opens boxes sequentially and can stop at any moment
- We will distinguish two options:
 - free recall – returning to previously rejected options is free
 - no recall – Pandora cannot go back to discarded options

SeqS: Preliminaries

Simple setup:

- two boxes
- no discounting
- free recall

Suppose that the first box has been opened. How should Pandora proceed?

Intuitively, high prizes should be accepted immediately (save on the search cost), and the prize is low, then it is better to search:

$u_1 \geq z \Rightarrow \text{stop}$

$u_1 < z \Rightarrow \text{search}$

z is called the **reservation value**

SeqS: Reservation Value

As Pandora is indifferent between accepting a box with prize z and continuing to search, it must be the case that

$$\begin{aligned} z &= \mathbb{E} \max\{z, u\} - s \\ &= -s + \Pr(u < z)z + \Pr(u \geq z)\mathbb{E}(u|u \geq z) \\ &= -s + F(z)z + [1 - F(z)] \int_z^{\bar{u}} u \frac{f(u)}{1 - F(z)} du \\ &= -s + z - [1 - F(z)]z + \int_z^{\bar{u}} u dF(u) \\ &= -s + z + \int_z^{\bar{u}} (u - z) dF(u) \end{aligned}$$

SeqS: Gains from Search

Thus, the reservation value is defined by

$$s = H(z) = \int_z^{\bar{u}} (u - z) dF(u) = \mathbb{E} \max\{u - z, 0\}$$

where $H(z)$ are gains from search

- $H(z)$ non-negative
- $H(z)$ is decreasing

$$\frac{\partial H}{\partial z} = -[1 - F(z)]$$

- $H(z)$ is convex

$$\frac{\partial^2 H}{\partial z^2} = f(z)$$

- $\lim_{z \rightarrow \bar{u}} H(z) = 0$
- $\lim_{z \rightarrow \underline{u}} H(z) = \mathbb{E}u$

Thus, the interior solution exists if $s < \mathbb{E}u$ and is unique

SeqS: $T = \infty$, Setup

We start with free recall and $\delta = 1$

Try to formulate the model in a similar style to our dynamic optimization lecture

State variable:

$$\hat{u}_t = \max\{u_1, \dots, u_t\}$$

Let $V_t(\hat{u}_t)$ be the value function of our Bellman Equation

Reservation value is the value of the best option at which Pandora is indifferent, so

$$z_t = V_t(z_t)$$

SeqS: $T = \infty$, Optimal Stopping

We look at the expected value from searching $t + 1$ boxes:

$$\begin{aligned}
 \mathbb{E}(V_{t+1}|\hat{u}_t) &= -(t+1)s + \int_{\underline{u}}^{\bar{u}} \max\{\hat{u}_t, u\} dF(u) \\
 &= -(t+1)s + \int_{\underline{u}}^{\hat{u}_t} \hat{u}_t dF(u) + \int_{\hat{u}_t}^{\bar{u}} u dF(u) \\
 &= -(t+1)s + \hat{u}_t F(\hat{u}_t) + \int_{\hat{u}_t}^{\bar{u}} u dF(u) \\
 &= -(t+1)s + \hat{u}_t + \int_{\hat{u}_t}^{\bar{u}} (u - \hat{u}_t) dF(u) \\
 &= -(t+1)s + \hat{u}_t + H(\hat{u}_t) \\
 &= V_t(\hat{u}_t) + [H(\hat{u}_t) - s]
 \end{aligned}$$

Thus, Pandora only gains from searching if $H(\hat{u}_t) - s > 0$.

SeqS: $T = \infty$, Optimal Rule

Thus, for all t we have

$$H(z_t) = s$$

all reservation values are the same $z_1 = z_2 = \dots = z$ and the same as in two-period model

Note, that as recall never happens on the optimal search path, the reservation value without recall is exactly the same

Probability of stopping is $1 - F(z)$

Thus, the expected duration of search is

$$\tau^e = \frac{1}{1 - F(z)}$$

SeqS: $T = \infty$, Adding Discounting

$$\begin{aligned} z &= \delta [\mathbb{E} \max\{z, u\} - s] \\ &= \delta \left[-s + zF(z) + \int_z^{\bar{u}} u dF(u) \right] \\ &= \delta \left[-s + z + \int_z^{\bar{u}} (u - z) dF(u) \right] \\ &= \delta [H(z) - s + z] \end{aligned}$$

Thus, we get the following equation for z :

$$s = H(z) - \frac{1 - \delta}{\delta} z$$

SeqS: Finite T , No Recall, Prelims

Bellman Equation:

$$V_t(x) = \max \left\{ x, -s + \int_{\underline{u}}^{\bar{u}} V_{t+1}(u) dF(u) \right\}$$

with boundary condition

$$V_T(u) = u$$

Claim

$$z_1 \geq z_2 \geq \dots z_T = 0$$

Proof. Let $z_{t+1}, \dots, z_T$ be the optimal reservation values from period $t+1$ onwards

Then, in period t Pandora can commit to using the same reservation values shifted by one period and not opening box T

As this guarantees exactly the same expected payoff as in period $t+1$, we have $z_t \geq z_{t+1}$

SeqS: Finite T , No Recall, Reservation Values

$$\begin{aligned}
 z_t &= -s + \int_{\underline{u}}^{\bar{u}} V_{t+1}(u) dF(u) \\
 &= -s + \int_{\underline{u}}^{\bar{u}} \max \left\{ u, -s + \int_{\underline{u}}^{\bar{u}} V_{t+2}(u) dF(u) \right\} dF(u) \\
 &= -s + \int_{\underline{u}}^{\bar{u}} \max \{ u, z_{t+1} \} dF(u) \\
 &= -s + z_{t+1} + \int_{z_{t+1}}^{\bar{u}} (u - z_{t+1}) dF(u) \\
 &= z_{t+1} + [H(z_{t+1}) - s]
 \end{aligned}$$

Suppose that $z_{t+1} > z$, the infinite horizon value. Then, $H(z_{t+1}) < s \Rightarrow z_t < z_{t+1} \Rightarrow$ contradiction

Sequence $\{z_t\}_{t=1}^T$ is bounded above by z and as $T \rightarrow \infty$ all the values converge to z

SeqS: Finite T , Free Recall

We can use similar arguments to state that $z_1 \geq \dots z_T$ and $V_1(u) \geq \dots V_T(u)$

That means, that if a Pandora is indifferent between accepting or rejecting $u = z_t$, then this u will be accepted in the next round as $u \geq z_{t+1}$

Thus,

$$z_t = -s + z_t F(z_t) + \int_{z_t}^{\bar{u}} u dF(u) = z_t + [H(z_t) - s]$$

Thus, the reservation value is **stationary and the same as in the infinite horizon setting**:

$$H(z) = s$$

SeqS: Remarks

- We considered the case when each search brings just one offer, sometimes it is a random number, see Lippman and McCall (1976) for details
- Sometimes past offers disappear with certain probability, see Karni and Schwartz (1977)
- Return can be costly, see Janssen and Parakhonyak (2014)
- Distribution might not be known, see next slide based on Rothschild (1974)

Rothschild (1974) in a Nutshell

Consider Pandora searching in an environment where **price** is:

- either 15 or 10 (with probabilities 0.9 and 0.1);
- or 10 or 5 (with probabilities 0.5 and 0.5);
- both these distributions are ex ante equally likely.

What is the optimal stopping rule if the search cost is 1?

- If $p = 5$ is observed, then clear that Pandora has to stop.
- If $p=15$ is observed, Pandora knows next search either yields 15 or 10. The expected cost in the case of continuation is

$$15 \times 0.9 + 10 \times 0.1 + 1 = 15.5 > 15 \Rightarrow \text{stop!}$$

- If $p=10$ is observed, the consumer has to update beliefs about which environment: it is five times more likely prices follow lower distribution:
 - $Pr(dist1|p = 10) = \frac{0.1 \times 0.5}{0.1 \times 0.5 + 0.5 \times 0.5} = \frac{1}{6}$.
 - Then, the cost of searching is:

$$\frac{1}{6}(15 \times 0.9 + 10 \times 0.1) + \frac{5}{6}(10 \times 0.5 + 5 \times 0.5) + 1 = 9\frac{2}{3} < 10$$

- Thus, it is optimal to continue.

1 Fixed Sample Search

2 Sequential Search

3 **Ordered Search**

Ordered Search: Introduction

So far search was random: boxes were symmetric and Pandora was not directing their search towards specific boxes

What if we relax this symmetry assumption?

Heterogeneity can have two sources:

- 1 different search costs s_i for each option
- 2 different distributions $F_i(u)$

Now instead of just optimal stopping problem searchers also face selection problem

This setting was described as “Pandora problem” in a famous paper by Weitzman (1979)

I will follow an elegant discrete choice representation of this problem by Armstrong (2017)

Ordered Search: Pandora Rule

Suppose that we have n options, each characterised by a pair (s_i, F_i)

Assume free recall and 0 outside option

For each option i define **reservation value** z_i :

$$\mathbb{E}_i \max\{u - z_i, 0\} = s_i$$

Theorem (Pandora Rule)

Optimal search is defined by:

- ❶ **Selection Rule.** *If another box is to be opened, it should be the unopened box with the highest reservation value z_i*
- ❷ **Stopping Rule.** *Terminate search whenever the highest discovered payoff exceeds the maximum reservation value of unopened boxes*

Ordered Search: PR and Discrete Choice

Following Armstrong (2017) define:

$$w_i \equiv \min\{z_i, u_i\}$$

Note, that u_i is *realisation* of random variable

Claim

Pandora rule is equivalent to choosing the box with the highest w_i

Note, that $w_i > w_j$ implies one of three options:

- ① $z_i > z_j$ and $u_i > \min\{z_j, u_j\}$; so i is inspected first (as $z_i > z_j$) and chosen (as $u_i > z_j$ or $u_i > u_j$)
- ② $z_i < z_j$ and $u_j < \min\{z_i, u_i\}$; so j is opened first, but then i is opened (as $u_j < z_i$) and chosen (as $u_j < u_i$)
- ③ $z_i = z_j = z$ and $u_j < \min\{z, u_i\}$; so either just i or both boxes are opened, but i is chosen

Ordered Search: PR and Discrete Choice

The claim shows choice equivalence between Pandora Problem and the following hypothetical situation:

- 1 searcher opens all boxes for free and observes realisations of u_i
- 2 then chooses the box with the largest w_i

Example. Let $(z_1, u_1) = (4, 2)$, $(z_2, u_2) = (5, 3)$, $(z_3, u_3) = (2, 6)$

Note, that $w_1 = 2$, $w_2 = 3$, $w_3 = 2$, so option two is chosen

Check Pandora Rule:

- 1 as $z_2 > \max\{z_1, z_3\}$, open box 2 first
- 2 as $u_2 < z_1$, open box 1
- 3 as $u_2 > z_3$ stop and take box 2

Ordered Search: Proof of Pandora Rule

Let $\mathbb{A}_i \in \{0, 1\}$ be selection indicator (with $\sum \mathbb{A}_i \leq 1$) and $\mathbb{I}_i \in \{0, 1\}$ be search indicator function. Then, Pandora's expected payoff is

$$\begin{aligned}\mathbb{E} \left(\sum \mathbb{A}_i u_i - \sum \mathbb{I}_i s_i \right) &= \mathbb{E} \left(\sum \mathbb{A}_i u_i - \sum \mathbb{I}_i \max\{u_i - z_i, 0\} \right) \\ &= \mathbb{E} \left(\sum \mathbb{A}_i [u_i - \max\{u_i - z_i, 0\}] \right) \\ &= \mathbb{E} \left(\sum \mathbb{A}_i \min\{u_i, z_i\} \right) = \mathbb{E} \left(\sum \mathbb{A}_i w_i \right) \\ &= \mathbb{E} (\max\{w_i\})\end{aligned}$$

Note that if Pandora opens a box i , but does not select it, i.e. $\mathbb{A}_i < \mathbb{I}_i$, it must be the case $u_i \leq z_i$

Note, that Pandora rule selects the box with highest w_i

Thus, Pandora rule delivers the highest possible payoff